Mathematical Statistics

Introduction

Xiao-Hua Zhou

Peking University

September 8, 2025: 3:10-5:00pm, Science Teaching Building 407 (理教 407)

Syllabus, 2025 Autumn Semester

- Lecturer: Xiao-Hua Zhou (azhou@math.pku.edu.cn), PKU Endowed Chair Professor, Peking University
- Grade: Final (50%), homework (20%), and mid-term (30%)
- Textbook books:
  - G. Casella and Roger L. Berger Statistical Inference. Second edition. Thomson Learning, 2002.
  - G. W. Imbens and D.R. Rubin (2017). Causal Inference for Statistics, Social, and Biomedical Sciences. Cambridge University Press.
- Reference books:
  - J. Rice. Mathematical Statistics and Data Analysis, 3rd Edition. Duxbury Advance Series. Thomson Learning, 2007
  - L. Wasserman. All of Statistics. A Concise Course in Statistics Inference. Springer, 2004

Content of lectures

- Introduction to Mathematical Statistics
- Principles of Data Reduction
- Estimation Theory
- Hypothesis Testing
- Confidence Sets (Intervals)
- Analysis of Variance
- Linear Models and Regression
- Statistical Decision Making and Bayes Statistics
- Causal Inference

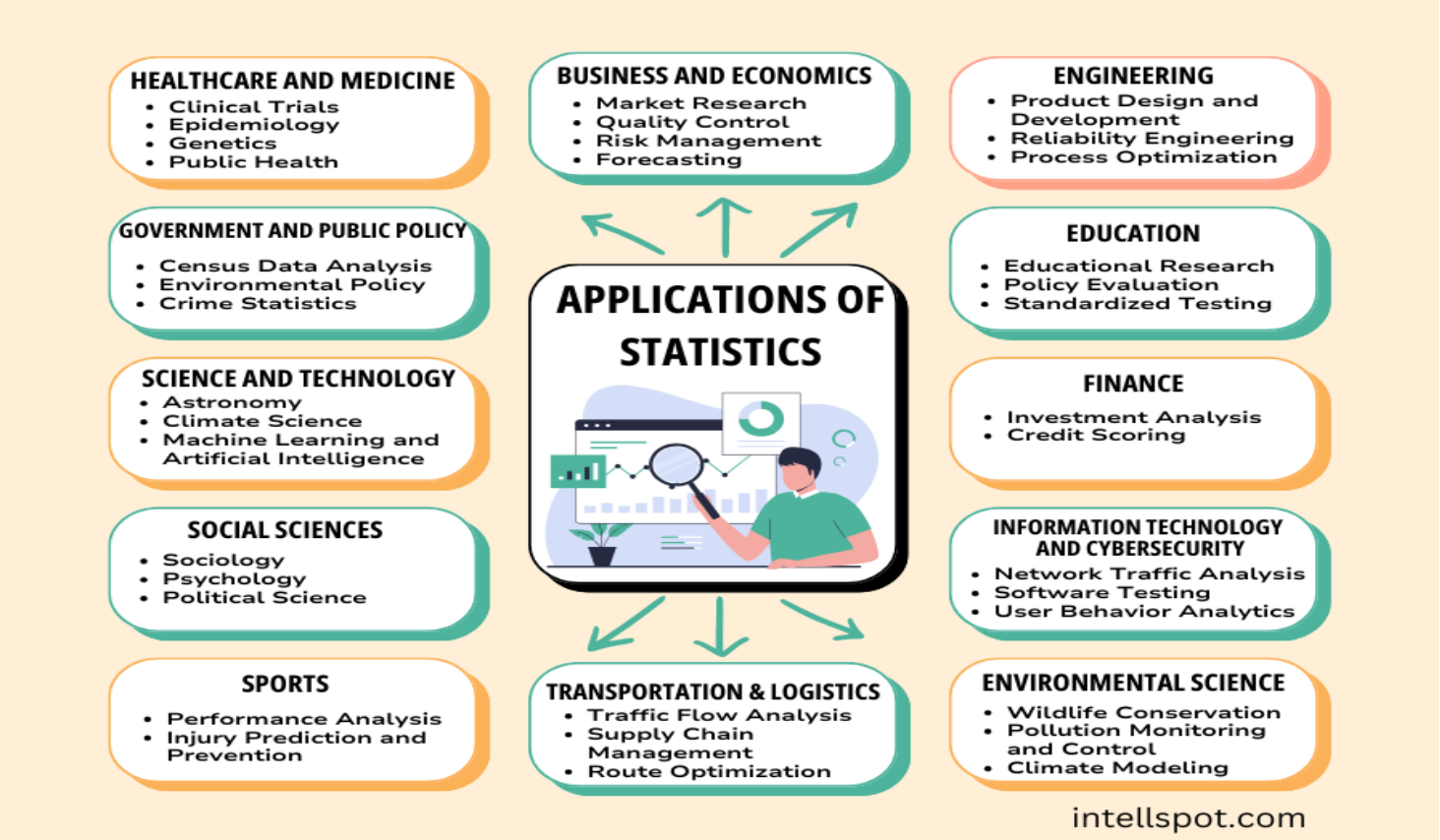
Goals of Learning Statistics

- Understanding Everyday Numbers: numbers are everywhere and every time
- Making Informed Decisions: make better decisions in your daily life, such as choosing a new phone, choosing a college major, or which movie to watch,
- Building Critical Thinking Skills: building skills to think clearly and question information you have.

Building Critical Thinking Skills

- Critical thinking: intellectual process of actively and skillfully conceptualizing, applying, analyzing, synthesizing, and evaluating information gathered (Scriven and Paul (2003))
- Need to have skills to ask questions like:
  - Where did these numbers come from?
  - How many people did they ask or measure?
  - Is there any chance the data is biased?

Some Real-Life Examples



Medicine (Biostatistics)

- Statistics are behind every medical study you hear about
- Statistics is used in discovering risk factors of a disease
- Statistically valid results have to be shown on the safety and effectiveness of a new drug before the drug can be approved by FDD or CFDA:
  - Randomized Controlled Trials (RCTs): Patients are randomly assigned to a treatment or placebo group.
  - Statistical Analysis: Researchers compare recovery rates using p-values (probability that results happened by chance).

Florence Nightingale (1820-1910): The Statistician and Nurse

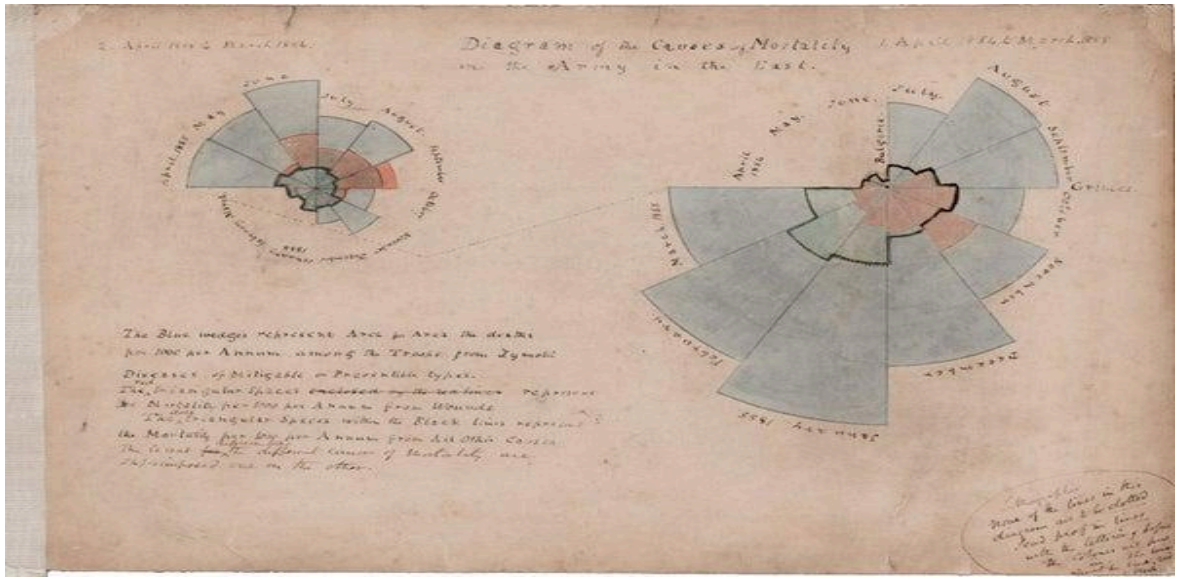


Crimean War (1854-1856): Recorded soldier mortality data and proved poor sanitation, not battle wounds, caused most deaths.

Statistical Contribution and Modern Nursing

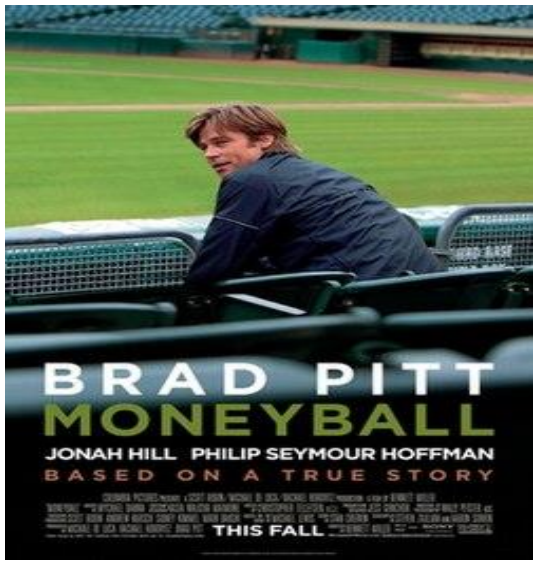
- Statistical Tool: Improving hygiene (handwashing, clean water) reduced death rates from 42% $\rightarrow$ 2%.
- Used Data to Reform Hospitals:
  - Collecting statistics in Crimea, counting the number of soldiers killed, injured or diseased—in the same way biologists collected specimens of butterflies and fossils on field trips.
  - Developing the modern hospital hygiene protocols.
  - Building up a team of data-gatherers at the Crimea.

Polar Area Diagram for Causal Inference



- Right side: The first year (1854–5), following her arrival in the region. Left: Second year (1855-6), after her intervention.
- Each wedge=a month, the area of the wedge= the number of soldiers who had died that month. The blue area= deaths from preventable diseases due to the terrible conditions at the Crimea. Red sections=deaths from battlefield wounds. Black areas=deaths from other causes.

Application in Sport: Moneyball



- Moneyball is a 2011 American biographical sports drama film.
- The general manager Billy Beane of the Oakland Athletics baseball team in the 2002 season assembled a competitive team with half (or even a third) as much money as his rivals by using a sophisticated sabermetric approach, the work of Bill James, a statistician who revolutionized the analysis of baseball statistics, to scouting and analyzing players.

Statistical Methods behind Moneyball

- Traditional baseball scouting relied heavily on subjective evaluations of players' physical attributes and intangible qualities. Teams often focused on metrics like batting average and RBIs for hitters, and wins and ERA for pitchers.
- Billy Beane sought out players overlooked by traditional scouting methods. He targeted athletes with specific skills that contributed to winning games but were underappreciated by the market.
- Key sabermetric concepts included: Weighted On-Base Average (wOBA), Fielding Independent Pitching (FIP), Wins Above Replacement (WAR)

Consumer Goods



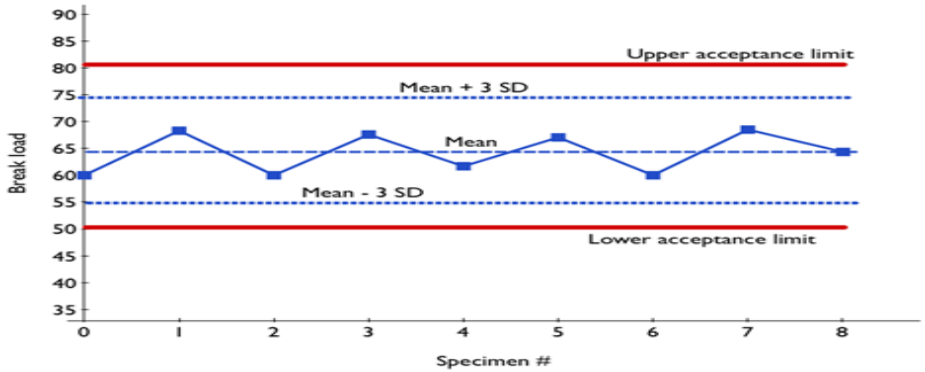
Consumer Goods

- From the analysis of their vast information, Walmart can decide which store stacks which product.
- For example, if Wal-Mart decided that people buy strawberry Pop Tarts when a hurricane is predicted in Florida, they ship this product to Florida stores based upon the weather forecast.



Quality Testing

- Companies make thousands of products every day and each company must make sure that each product has good quality.
- But a company can't test each and every item that they produce.
- So the company uses statistics to test just a few, called a sample, of what they make.
- If the sample passes quality tests, then the company assumes that all the items made in the group, called a batch, are good.



What is Statistics

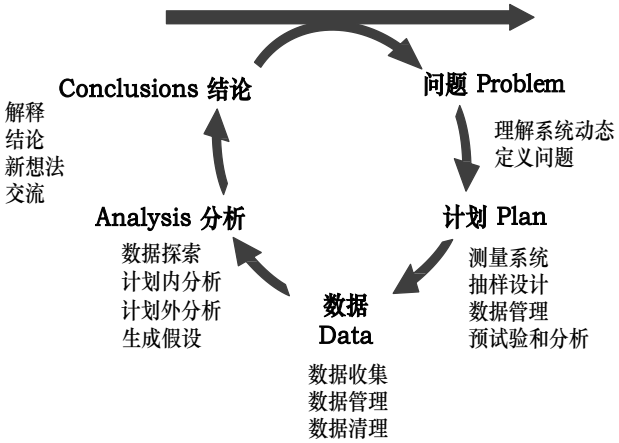
- Statistics is the science of learning from data, and of measuring, controlling, and communicating uncertainty; and it thereby provides the navigation essential for controlling the course of scientific and societal advances (Davidian and Louis (2012), Science).
- This field will become ever more critical as academia, businesses, and governments rely increasingly on data-driven decisions, expanding the demand for statistics expertise.

Difference between Mathematics and Statistics

- 统计学尽管源于数学，却是一门方法论学科，它并不局限于自身发展，而是致力于为其他领域提供一套连贯的数据处理思想和工具 (Cobb (1997))
- 为了实现该目标，统计学需要以数学作为工具。
- 在数学中，研究者对数学结构本身感兴趣
- 而在统计学中，数学结构只是被用作发展统计理论从而解决现实世界问题的工具。
- 因此，数学学家倾向于获得近似问题的确切答案，而应用统计学家更倾向于获得确切问题的近似答案。

统计思维的四个维度

维度1：研究周期



维度2：思维方式

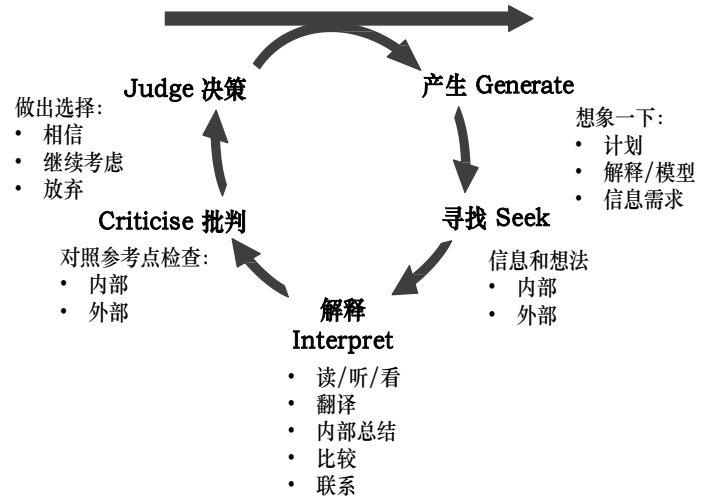
一般类型

- 策略性
  - ——计划，预知问题
  - ——对现实约束条件的认识
- 寻求解释
- 建立模型
  - ——先建立后使用
- 应用方法
  - ——遵循先例
  - ——问题原型的识别和使用
  - ——使用解决问题的工具

统计思维的基本类型（基础）

- 认识到需要数据
- 数据分析（改变表现形式以促进理解）
  - ——从真实系统中捕捉“测度”
  - ——改变数据表现形式
  - ——数据中传递信息
- 考虑变异
  - ——发现和确认
  - ——为了预测、解释和控制而进行测量和建模
  - ——解释和处理
  - ——研究的策略
- 用统计模型推理
  - ——基于聚合的推理
- 将统计与实际情境相联系
  - ——信息、知识和概念

维度3：疑问周期



维度4：性格特质

- 怀疑精神
- 想象力
- 好奇心和觉察力
  - ——善于观察、注意
- 接受力强
  - ——对于与既往观念冲突的想法
- 探究深层意义的倾向
- 逻辑性
- 参与度
- 毅力

Mis-use of An example on Air Plane



- During World War II, the Royal Air Force wanted to add more armor to their planes, but because of weight limits, they needed to know which places needed the armor most.

An Example on Air Plane



- So, they examined the planes after they came back and counted how often bullet holes were found in certain areas and then placed armor in places that showed the fewest bullet holes.
- This is because, they assumed, that any place that did have bullet holes was a place that planes could be hit and still fly.
- Problem: No plane that ever came back had holes where the gas tank was. Because planes whose tank was hit would explode and not come back - biased sample.

Another Example on Polio and Food



- During the 1952 pandemic 58,000 American children were infected. More than three thousand died with over 20,000 left paralyzed.

Another Example on Polio and Food

- Prior to the polio vaccination success there were many public policy attempts to try to prevent the spread of this disease.
- When ice cream consumption increased the polio disease increased. The Public Health Office in North Carolina (and other states) issued warnings against eating ice cream.

Reasons

- The polio virus became more active during hot weather.
- The same hot weather also drove the consumption of ice cream.
- But ice cream did not cause polio nor did it help the spread of polio.
- Correlation does not imply causality.

Smallpox (小儿麻痹症) and Vaccine (疫苗)



- 1916 年第一次小儿麻痹症 (Polio) 流行病袭击了美国, 成千上万的受害者.
- 1950 年 Jonas Edward Salk 培育了一种小儿麻痹症疫苗.

小儿麻痹症的现场试验

- 1954 年对小学 1, 2, 3,4, 5 年级儿童进行试验.
- 效果的评价方法: 新药的效果如何? 试验和比较

若干种方案设计方案

1. 给大部分儿童接种, 无对照组; (可能每年流行程度不同, 难与前一年对比)
2. 父母同意的儿童接种, 不同意的不接种; (可能高收入的父母比低收入的容易同意)
3. 对所有 2 年级儿童接种, 1 和 3 年级作为对照; (本病是接触传染, 容易有偏差)
4. 根据人的判断, 使对照组与处理组有相同的背景变量的值, 如: 年龄, 收入, 健康, 习惯等; (但判断难免偏倚, 仍可能有未观测的混杂因素)
5. 抛硬币, 进行随机分组, 即随机化对照试验. 机会法则保证两组相近.

随机对照双盲试验

- 随机化试验方法: 将病人随机地分到处理组和对照组.
- 双盲:
  - 对病人: 采用 “安慰剂”;
  - 对医生: 不知道病人是否接受处理.

随机对照双盲试验

	人数	比率
处理	200,000	28/10 万
对照	200,000	71/10 万
不同意	350,000	46/10 万

疫苗效果: $(71-28)/10^5 = 43/10^5$.

小儿麻痹基金会 (NFIP) 研究

National Foundation for Infantile Paralysis (NFIP)

小儿麻痹基金会 (NFIP) 研究

	人数	比率
2 年级 (处理组)	225,000	25/10 万
1,3 年级 (非随机安慰对照组)	725,000	54/10 万
2 年级 (不同意)	350,000	44/10 万

疫苗效果: $(54-25)/10^5 = 29/10^5$.

- NFIP 的研究过低评价了接种的效果.
- 偏性来源于混杂.
- NFIP 的处理组不含父母不同意的儿童;
- NFIP 的对照组含父母不同意的儿童.

统计学的发展历史

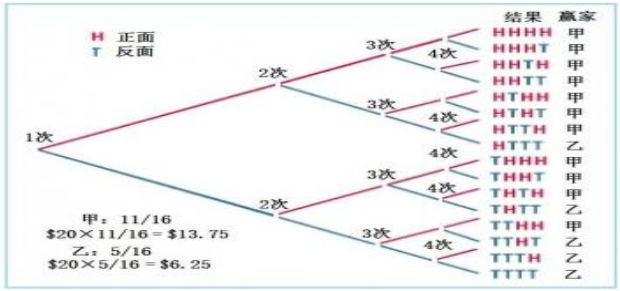
- 统计科学的发展总是和真实世界中的问题息息相关，比如博弈问题、物理研究、社会科学、生物学、农业、生物医学等等。
- Fienberg (1992) 和 Mathews (2016) 把统计学科的历史分为如下几个阶段:
  - (1) 萌芽阶段（史前阶段），即 1654-1750 年; 这一阶段中以博弈中的一些问题为出发点推动了统计学科的建立。
  - (2) 基本统计方法与物理尤其是天文学的结合发展阶段即 1750-1820 年.
  - (3) 基本统计方法与社会科学和生物学的结合发展阶段，即 1820-1900 年.
  - (4) 现代统计方法学阶段，即 1900-1950 年，此时统计学常与农业中的问题相结合。

十八世纪前的发展历史:
经典概率论

1660-1708：概率论的诞生 Blaise Pascal 和 Pierre de Fermat:

- In 1654 年，梅雷 (Chevalier de Mere) 向帕斯卡 (Blaise Pascal) 请教了一个“分赌注问题”：赌友 A 和赌友 B 各自出 32 枚金币，共 64 枚金币作为赌注，双方以掷骰子为赌博方式：如果结果出现“6”，则赌友 A 赢 1 分；如果结果出现“4”，则赌友 B 赢 1 分。双方谁先得到 10 分，谁就赢得全部赌注。
- 赌博进行了一段时间后，赌友 A 已得 8 分，赌友 B 也得了 7 分。但这时，赌友 A 有事，中断赌博。Q：这 64 枚金币的赌注应该如何分配才合理呢?

Blaise Pascal 和 Pierre de Fermat



（a）费马列出所有结果计算分配比例

结果				概率	所得（甲）	概率加权所得
H	H			1/4	20	5
H	T	H		1/8	20	5/2
H	T	T	H	1/16	20	5/4
H	T	T	T	1/16	0	0
T	H	H		1/8	20	5/2
T	H	T	H	1/16	20	5/4
T	H	T	T	1/16	0	0
T	T	H	H	1/16	20	5/4
T	T	H	T	1/16	0	0
T	T	T		1/8	0	0

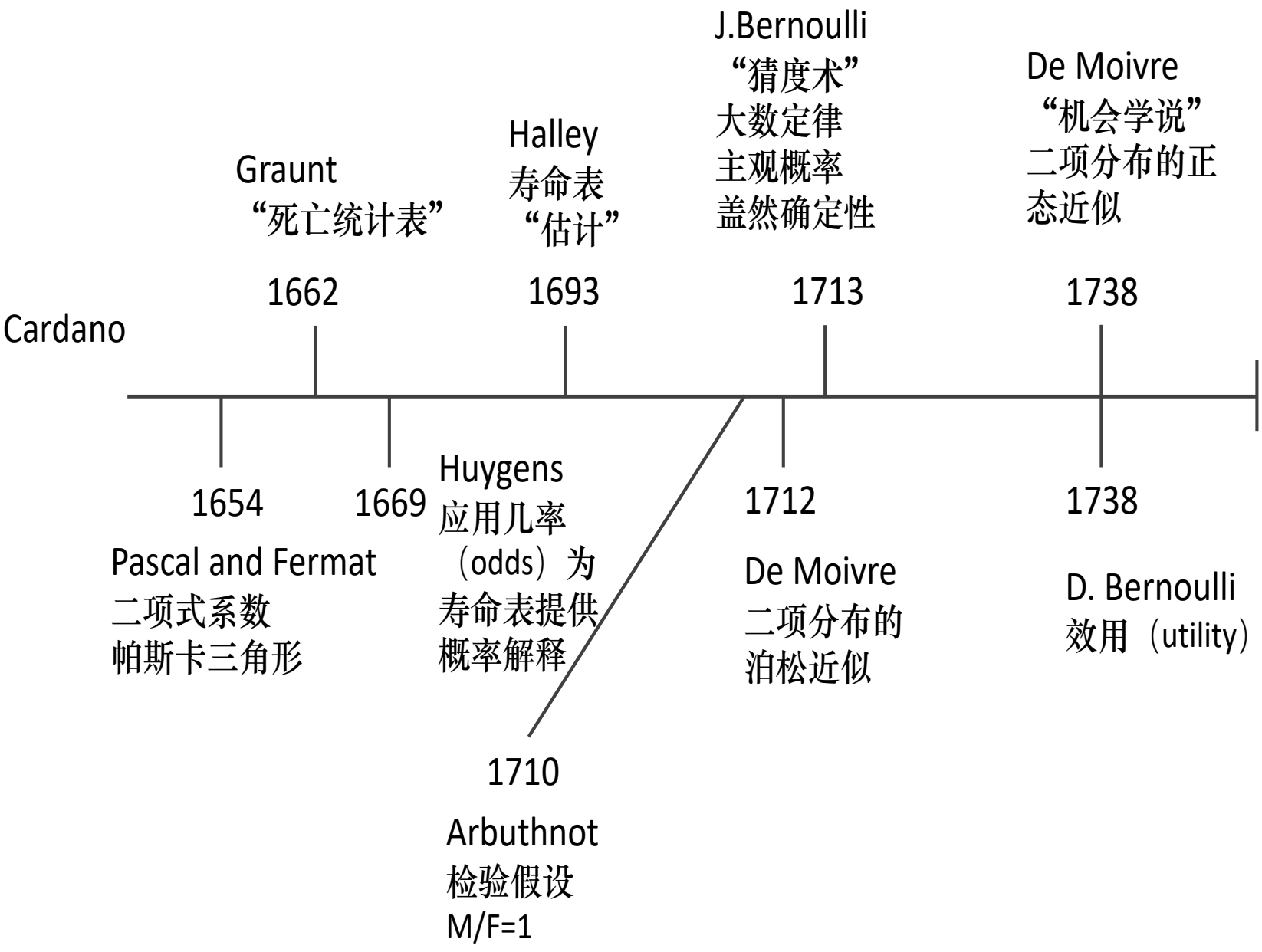
期望值（甲）55/4 = $13.75

（b）帕斯卡引入期望值的概念计算所得（甲）

原文链接：https://blog.csdn.net/weixin_42428226/article/details/120096859

- 费马 solution：中断赌博时赌友 A 得了 8 分，距离 10 分的判据差 2 分，赌友 B7 分，还差 3 分到 10 分。费马列出了“梅雷问题”中赌博继续下去的各种结果。经过 4 次随机抛丢或产生 16 种等概率的可能结果，如上图 (a) 中最右侧所列. 算出甲赢的概率是 11/16，赢的钱是 $13.75；乙赢的概率是 5/16，赢的钱是 $6.25
- 帕斯卡 solution: 提出了离散随机变量“期望值”的概念。期望值是用概率加权后得到的“期望”的平均值。如上图 (b) 所示，帕斯卡计算出从甲方的观点，“期望”能得到的赌注分配为 $13.75。

第一阶段：1654-1750



十九世纪的发展历史
物理中的基本统计学方法

- 1750-1820 是数理统计学科的开端时期，统筹被称为 Gauss-Laplace 时期
- 在这个时期，有两条相互交织的统计学发展线索
- 一条是用于估计线性方程中未知系数的最小二乘法的发展
- 另一条是由 Bernoulli 和 De Moivre 所做工作衍生出来的基于概率论的推断

十九世纪的发展历史
物理中的基本统计学方法

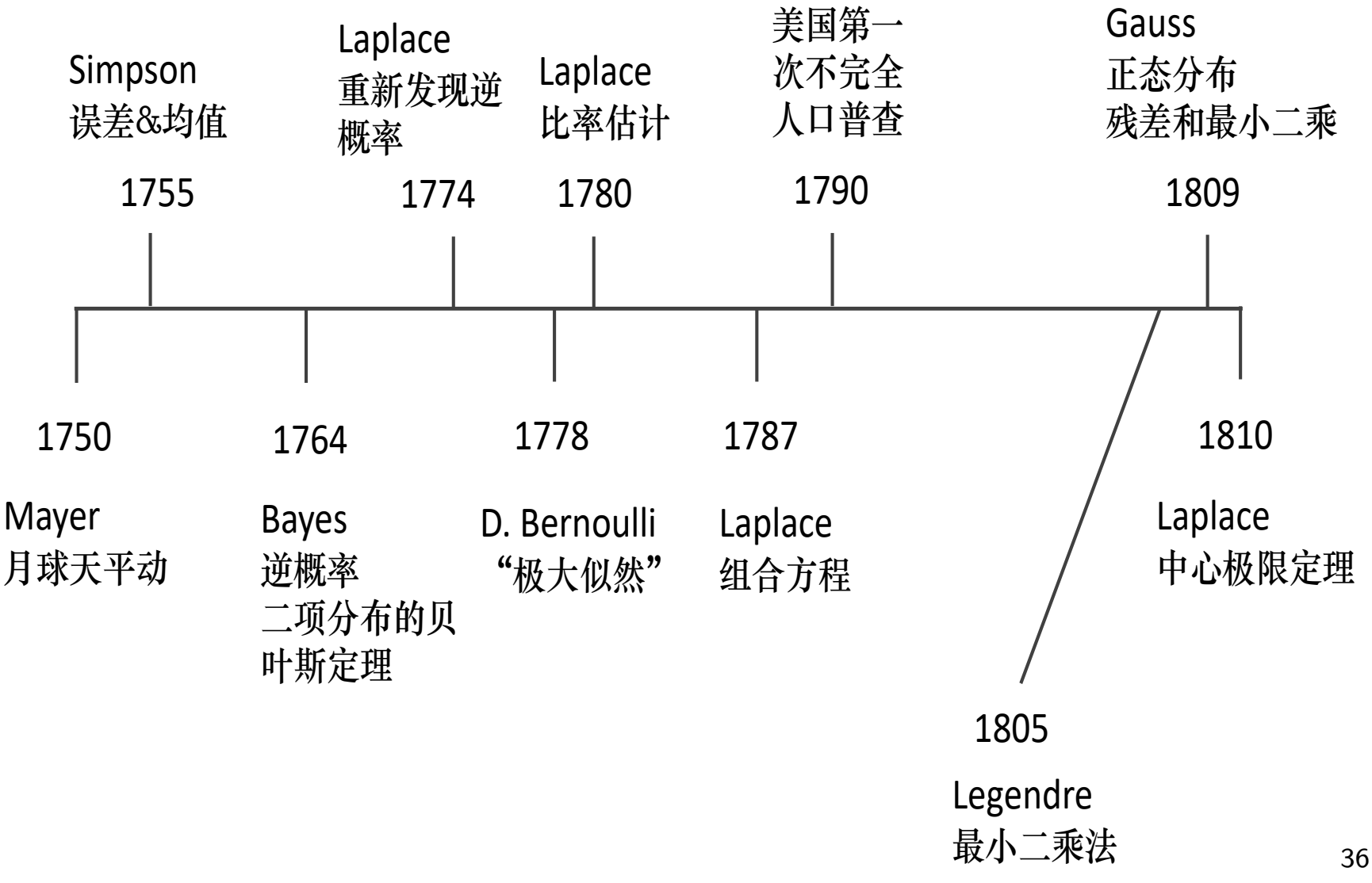
在这个时期有许多代表性的工作。

- Thomas Bayes 利用了逆概率公式发展了贝叶斯理论，并在数据的基础上对二项分布的参数 p 进行了估计
- 法国数学家 Pierre Simon Laplace 也利用了逆概率公式在 1810 年发展了中心极限定理理论，并证明了可以利用正态分布来近似任何类型概率分布的平均和
- 德国数学家 Carl Friedrich Gauss 利用了 Laplace 发展的基于概率论的推断方法给出了最小二乘法和不确定性量化间的关键联系
- 基于 Gauss-Laplace 时期理论的统计推断使用概率论进行论证，提供了由样本观测推广到大量群体的归纳过程

十九世纪的发展历史
物理中的基本统计学方法

- Bernoulli's Law of Large Numbers provided the theoretical justification for why statistics works in the long run. However, the exact calculations for a large number of trials were impossibly complex at the time.
- De Moivre provided the necessary computational tools by showing that the normal curve could serve as a powerful, manageable approximation, effectively solving the problem that had plagued Bernoulli.
- Together, their work transitioned the field of statistics from theoretical speculation to a practical, quantitative science.

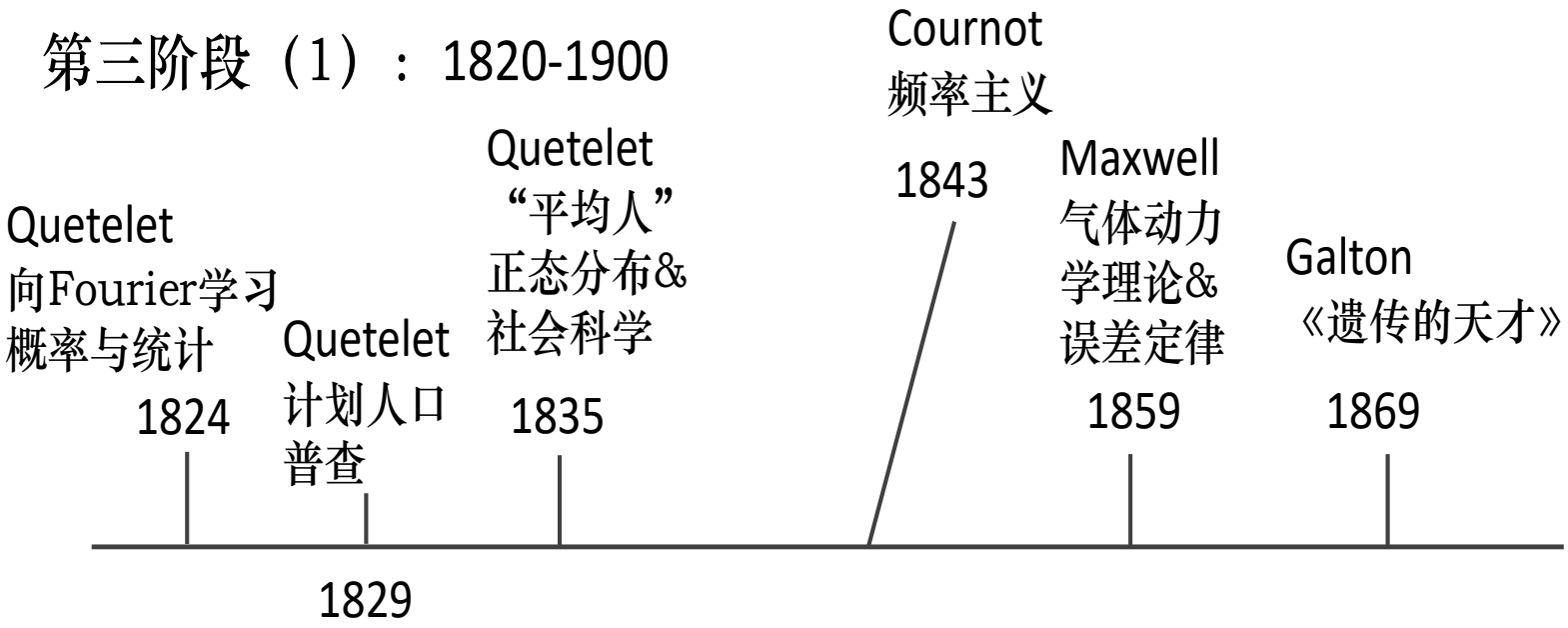
第二阶段：1750-1820



十九世纪的发展历史社会科学和生物学的基本统计方法

- 在社会科学问题的驱动下，Francis Galton 发展了回归的思想以及回归与正态分布的相关联系，并且引入了相关性的概念
- 之后 Francis Edgeworth 将 Galton 的回归和相关性的思想拓展到多元正态分布的背景下，并引入了相关性矩阵的记号
- 在生物学问题的驱动下，Karl Pearson 提出了偏态曲线的 Pearson 曲线族并针对列联表的拟合程度问题发展了卡方检验方法
- 为了在相关性理论和回归之间建立联系，Yule 发展了最小二乘法的早期方法和误差理论，并提出了复相关和偏相关的概念

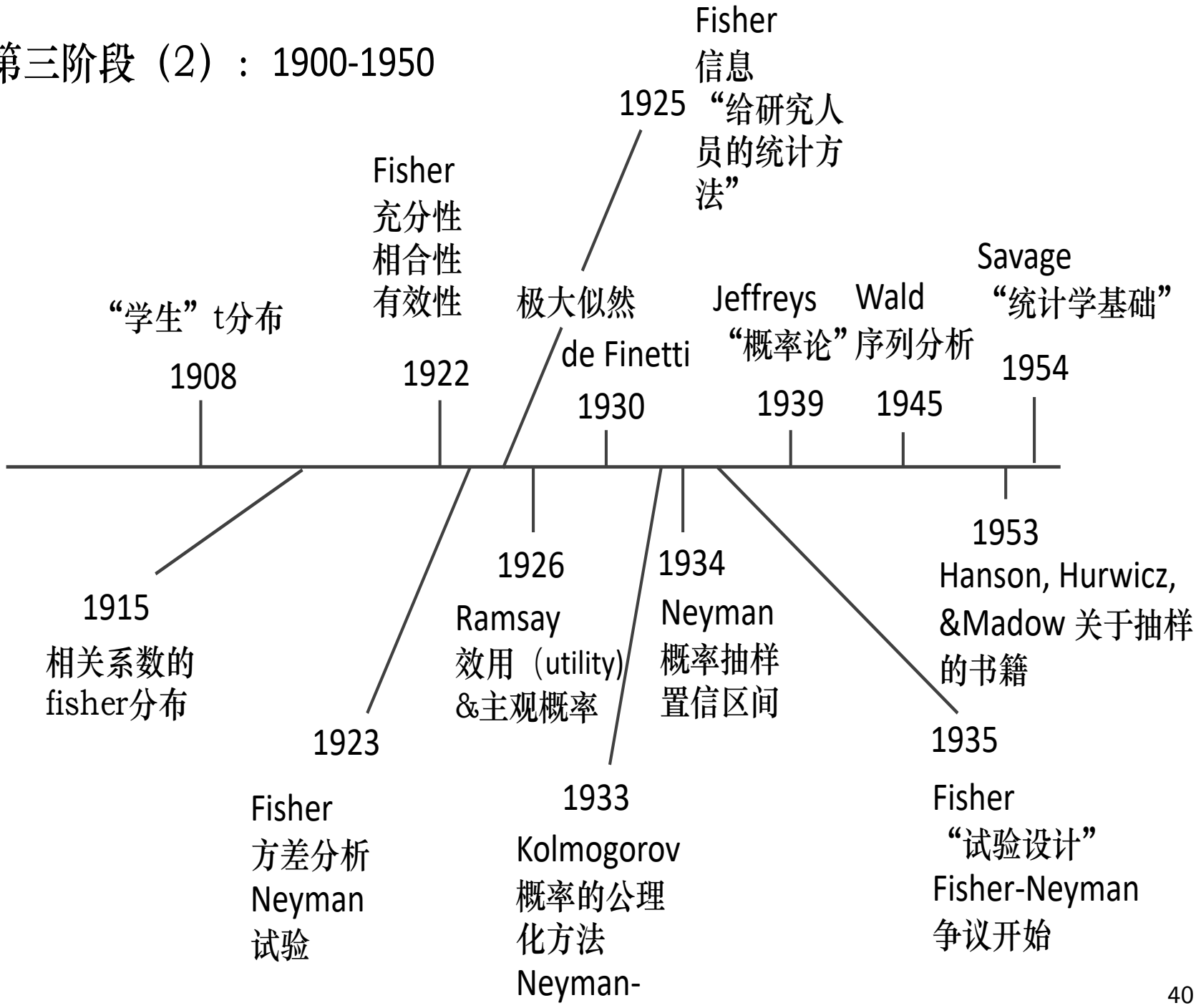
第三阶段（1）：1820-1900



二十世纪的发展历史
现代统计学方法

- 在农业领域问题的驱动下，英国统计学家 Ronald A. Fisher 发展了统计学模型符标记符号、充分性、可能性、随机化的统计概念、随机化意义下的实验设计理论、方差分析方法等相关理论波兰统计学家 Jerry Neyman 在他 1923 年的博士论文中明确地使用了假设性回答的相关记号回答了当处理分配方案不同时将会观测到什么，为基于潜在结果的因果推断打下了基础
- Jerry Neyman 和 Egon Pearson 也发展了假设检验的理论，他们认为能够通过明确地认清对立假设的角色来改进 Fisher 的统计学意义检验
- 在 1934 年，Jerry Neyman 为统计样本理论奠定了基础，并且基于重复样本的无限序列给出了关于置信区间估计方法的第一个描述

第三阶段（2）：1900-1950



现代临床试验的发展历史

- 《圣经》中的《但以理书》提到了最早的现代临床实验之一
- 巴比伦的尼布甲尼撒（Nebuchadnezzar）王想让他所有的臣民只吃肉喝酒，但是 Daniel 和其他犹太子民想要进食豆类和水
- 尼布甲尼撒王允许他们可以继续自己的饮食方式十天，之后如果他们比其他人更加健康就可以继续自己的习惯
- 最终，犹太人得到了保持自己饮食方式的许可

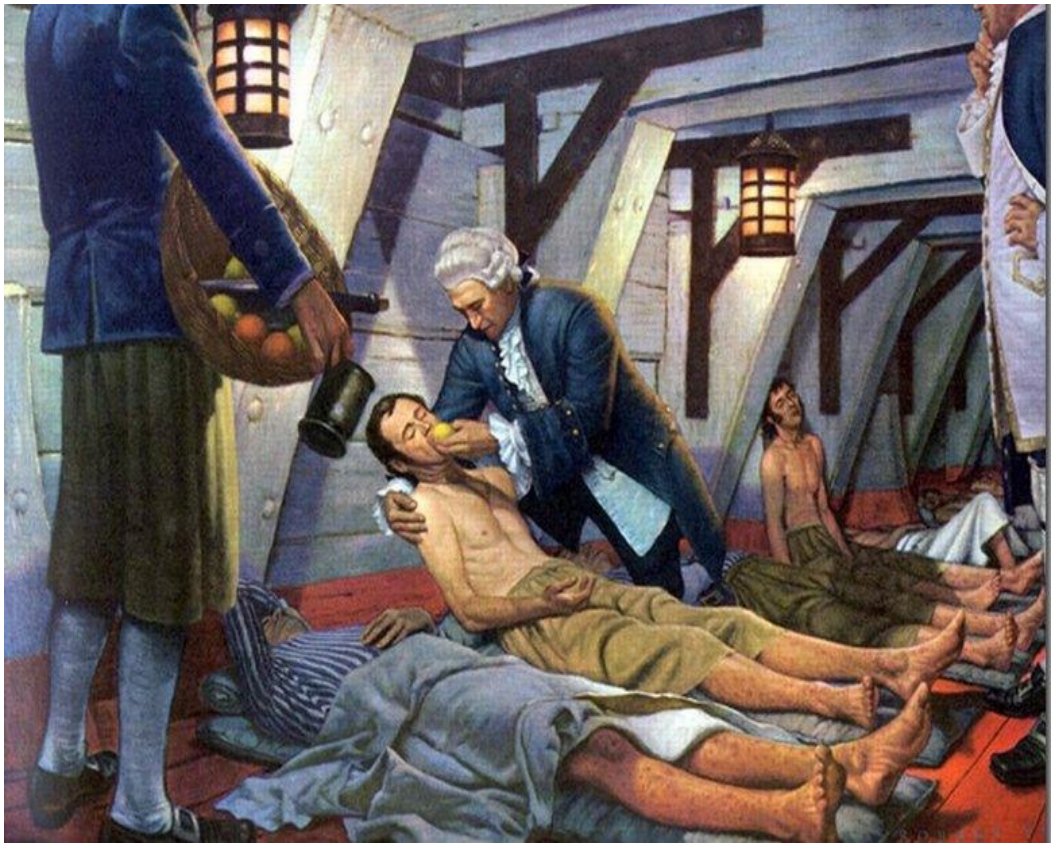


现代临床试验的发展历史

- James Lind (林德) (1765-1794)-一个随船外科医生——被称为控制临床实验之父
- 1757 年，林德被迫应对在水手中爆发的坏血病 (Scurvy)。他想检验他的使用特定饮食进行治疗的方法是否有效
- 在他的实验中，他检验了 6 种不同的主食以外的补充剂用于治疗坏血病，其中一种补充剂是每天两个橙子和一个柠檬 (富含维生素 C) (Two oranges and one lemon daily, which is rich in vitamin C). 其它是：(1) 每天一夸脱苹果酒 (A quart of cider a day), (2) 几滴硫酸 (Drops of sulphuric acid), (3) 每天半品脱海水 (Half a pint of sea-water a day), (4) 大蒜、芥菜籽、辣根、秘鲁香脂和没药树胶的混合物 (A mix of garlic, mustard seed, horse-radish, balsam of Peru and gum myrrh), (5) 两勺醋 (Two spoonfuls of vinegar)

现代临床试验的发展历史

- 选择了十二名症状相似的患者；都表现出“牙龈腐烂、斑点和倦怠，以及膝盖无力”。在参与试验期间，他们的饮食完全相同，包括稀粥、羊肉汤、清淡的布丁和煮熟的饼干。
- 他根据这十二名患者制定了不同的治疗方案；每两人接受相同治疗方案。



现代临床试验的发展历史

- 在一周的试验结束后，用餐中加入了橙子和酸橙的两名水手，其中一人完全恢复，另一人几乎康复，而其他五个小组中没有任何一个人病情有明显改善
- 他的试验被视为第一个众所周知的对照设计临床试验：一种平行对照组被给予一种替代疗法的试验
- 在试验结束后，用餐中加入了橙子和酸橙的两名水手，其中一人完全恢复，另一人几乎康复，而其他五个小组中没有任何一个人病情有改善
- 他的试验被视为第一个众所周知的对照设计临床试验：一种平行对照组被给予一种替代疗法的试验

现代临床试验的发展历史

- 19 世纪出现了使用安慰剂的试验
- 在医学上，安慰剂”是任何有意的无效的医学治疗，特别是无活性或者惰性物质，这些治疗用于替换那些病人希望接受的但不能或者不适合使用的疗法
- 安慰剂通常用于试验性研究中的对照控制组，与那些使用了研究用药的结果进行对照
- 在拉丁语中，安慰剂 Placebo 的字面意思是我愿意

现代临床试验统计方法

- 当 Galton 和 Pearson 在物理和生物学中发展基本统计方法的理论时，统计思想和推理也在 19 世纪的医学领域得到了应用
- 法国临床医生 Pierre-Charles-Alexandre Louis 是医学中使用数值方法的先驱。他认为，新的有效医学知识可以从汇总的临床数据中得出
- 在 19 世纪早期，各种各样的发热在欧洲很常见，而巴黎医生 Francois Joseph Victor Broussais，断言所有的发热都有相同的起因，并且都是人体器官炎症的表现。
- 他提出在发炎器官相对应的身体表面上施放水蛭 (leech) 的方法，将放血作为发热的有效治疗方法
- 当时的法国医生高度重视 Broussais 的有关发热的理论和治疗方法。

现代临床试验统计方法

- 而 Pierre-Charles-Alexandre Louis 对 Broussais 理论的有效性表示怀疑，并且在 1835 年，路易斯进行了一项临床研究来调查放血对肺炎患者的影响。
- 在这项研究中，他从大量的病例记录中选择了一组同质性较高的患者，他们有相同的、特征明确的肺炎型。
- 他在巴黎医院 LaCharite 进行了多年的临床实践把这些病例结合起来。在 Louis 的报告中，在 47 名进行放血的患者中有 18 名患者死亡，36 名未放血的患者中只有 9 名患者死亡，也就是死亡率更低。
- 路易斯进一步预测了数值推理（包括群体对比和基于人群的思维方式）将被广泛用于医学推断，而医学将不再是医生们滥用神学的自圆其说
- Louis 在那时的工作就已经预示着如今对循证医学的讨论重视

现代临床试验统计方法

- 然而，一些路易斯的批评者指出他的方法未能承认医生必须将个体而不是一个统计上的整体视为真正的病人（Alfredo Morabia（J R Soc Med 2006; 99：158-160））
- 因此，统计询证的意义在整个 19 世纪仍然存在争议
- 例如，著名的外科医生 Joseph Lister（1827-1912）主张基于统计学研究的特殊抗菌手术方法 (Lister’s evidence of the effects of antisepsis: case reports).
- Lister’s 消毒系统: (1) 外科医务人员人员-弱含酚的手洗液，(2) 器械-酚醇浴，(3) 病人- 使用酚醇喷雾溶液来降低患者周围空气中的细菌水平。
- 在 1870 年，Lister 在医学杂志《柳叶刀》上报告到他使用酚酸进行手术的结果。他记录了使用酚酸的影响：在使用酚酸之前，36 名患者中有 16 人死亡，死亡率为 46%。采用酚酸后，40 名患者中仅有 6 人死亡，死亡率降至 15%。

现代临床试验统计方法

- Karl Pearson 和 Ronald A. Fisher 最杰出的贡献是通过发展医学数值方法的标准化和数学理论来创建现代临床试验方法。
- Pearson 不仅发展了许多现代统计方法来研究生物变异以便明确达尔文自然选择理论的统计含义，而且还通过在《英国医学期刊》(British Medical Journal)、《柳叶刀》(Lancet)、《皇家医学期刊》(The Royal Society of Medicine) 等期刊上发表文章将这些方法拓展到医学界中，并教育医学专家使用正确的统计推理方法

现代临床试验统计方法

- 在 Pearson 发展生物数据统计方法时，Fisher 发展了他在农业领域的实验设计原则。
- Fisher 正确设计实验的主要思想包括：
  - (1) 随机化的重要性，也就是将个体随机分配给实验中的不同组
  - (2) 可重复性，测量和实验操作都应该可重复，以减少不确定性并确定变异来源
  - (3) 阻塞，将实验单元安排成彼此相似的单元组，以减少不相关的变异来源。
  - (4) 使用因子实验，可以有效地评估几个独立因素的影响和可能的相互作用 Fisher 提出的实验设计原则为现代临床试验奠定了基础，他的实验设计原则侧重于随机实验，
- 而 Sir Austin Bradford Hill 通过提出观察性研究的数据分析原则，进一步发展了临床试验方法。

Basic Concept in Mathematical Statistics (Statistical Inference)

- Population
- Sample
- Statistics
- Estimation
- Hypothesis Testing
- Decision Making

Population and Parameters

- The population in a study is the complete set of people or things being studied.
- The parameters are specific characteristics of the population that a statistical study is designed to estimate.

Random Variable and Random Sample

- Let X represent the measure of interest for a randomly chosen member from the population, such as the lifetime of an electronic device. Since we do not know in advance the value of X, X is random.
- Let $F(.)$ be the distribution of X and $f(.)$ be the corresponding density or mass function.
- We estimate F by a sample of observations from the population.
- The random variables $X_1, \ldots, X_n$ are called a random sample of size n from the population $f(x)$ if $X_1, \ldots, X_n$ are mutually independent random variables and the marginal pdf or pmf of each X_i is the same function $f(x)$.
- The joint pdf or pmf of $X_1, \ldots, X_n$ is given by

$$f(x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i).$$

Simple Random Sampling

- A simple random sample: a sample design in which every possible sample of size n from the population of N elements has an equal probability of selection.
- It may be selected by taking random draws from the set of numbers $\{1, 2, ..., N\}$.
- With simple random sampling, elements have equal probabilities of selection.

Random Sample from an Infinite Population

- The random sampling model above is also called sampling from an infinite population.
- Think of obtaining the values of $X_1, \ldots, X_n$ sequantially. First, the experiment is performed and $X_1 = x_1$ is observed.
- Then, the experiment is repeated and $X_2 = x_2$ is observed. The assumption of independence in random sampling implies that the probability for X_2 is unaffected by the fact that $X_1 = x_1$ was obtained first..
- "Removing"' x_1 from the infinite population does not change the population, so $X_2 = x_2$ is still a random observation from the same population.
- When the population is finite. the above model may not be true.

Random Sample from Finite Population I

- Simple random sampling with replacement (SRSWR): Suppose a value is chosen from the population in such a way that each of N values is equally likely (probability = 1/N) to be chosen. (Thinking of drawing a number from a hat). This number is recorded as $X_1 = x_1$.
- Then, the chosen number is returned to the population, and the process is repeated. Again, each of the N values is equally likely to be chosen. The second value chosen is recorded as $X_2 = x_2$.
- This process of drawing from the N values is repeated n times, yielding the sample $X_1, \ldots, X_n$.
- Standard statistical theory and analysis generally assumes SRSWR

Random Sample from Finite Population II

- The second method for drawing is called simple random sampling without replacement (SRSWOR), simple random sampling: once an element has been drawn, it is removed from the set of elements eligible for selection on subsequent draws.
- Assume that each of N $\{x_1, \ldots, x_N\}$ has equal probability of $1/N$ being chosen.
- The first value is chosen with the probability of $1/N$, and recored as $X_1 = x_1$.
- Now a second value is chosen from the remaining $N - 1$ value with equal probability of $1/(N - 1)$, and is recorded as $X_2 = x_2$.
- Choice of the remaining values continues in this way, yielding a sample $X_1, \ldots, X_n$.

Random Sample from Finite Population

- A sample drawn from a finite population without replacement is not a random sample. In fact, the random variables $X_1, \ldots, X_n$ are not mutually independent,
- However, when N is large, $X_1, \ldots, X_n$ are nearly independent.

Proof on Non-independence

- Let x and y be distinct elements of $x_1, \ldots, x_N$. Since the value y cannot be chosen at the second stage, $P(X_2 = y \mid X_1 = y) = 0$. However, $P(X_2 = y \mid X_1 = x) = 1/(N-1)$. Hence, the probability distribution for X_2 depends on the value of X_1 that is observed and hence, X_1 and X_2 are not independent.

Proof on Identically distributed

- $X_1, \ldots, X_n$ are identically distributed. That is, the marginal distribution of X_i is the same for each $i = 1, \ldots, n$.
- $$P(X_2 = x) = \sum_{i=1}^{N} P(X_2 = x \mid X_1 = x_i) P(X_1 = x_i).$$
- Note that for each value of the index, say, $x = x_k$, $P(X_2 = x \mid X_1 = x_k) = 0$. For all other $j \neq k$, $P(X_2 = x \mid X_1 = x_j) = 1/(N-1)$.
- Thus,
$$P(X_2 = x) = (N-1)(\frac{1}{N-1}\frac{1}{N}) = \frac{1}{N}.$$
- Similar arguments cane be used to show that each of the X_is has the same marginal distribution.

Statistics

- Let $X_1, \ldots, X_n$ be a random sample of size n from a population.
- Let $T(X_1, \ldots, X_n)$ be a real-value of vector-value function whose domain includes the sample space of $(X_1, \ldots, X_n)$.
- Then, the random variable or random vector $Y = T(X_1, \ldots, X_n)$ is called a statistic.
- The probability distribution of a statistic Y is called the sampling distribution of Y.
- Statistics is very broad but cannot be a function of parameters.

Sample and Sample Statistics

- A sample: a subset of the population from which data are actually obtained.
- Raw data: the actual measurements of observations collected from the sample.
- Sample statistics: characteristics of the sample that are derived from the raw data.

From Sample Statistics to Population Parameters

- Suppose Nielsen found that 10% of the people in the 5,000 homes in its sample watched the World Cup. This 10% is a sample statistics.
- What Nielsen really wants to know is the corresponding population parameter, the percentage of all Chinese who watched the World Cup.
- We can use sample statistics to estimate the population parameter. We may say that 10% of the Chinese watch the World Cup.
- How good is this estimate?

Estimation

- Let $X_1, \ldots, X_n \sim f(x; \theta)$.
- We want to estimate $\theta = (\theta_1, \ldots, \theta_k)$.
- We will discuss three methods of constructing estimators $\widehat{\theta} = w(X_1, \ldots, X_n)$:
  - The Method of Moment (MOM)
  - Maximum Likelihood (ML)
  - Bayesian Estimators.
- We will also discuss several methods for evaluating estimators:
  - Bias and Variance
  - Mean squared error (MSE)
  - Large sample theory

Hypothesis Testing

- Let $X_1, \ldots, X_n \sim f(x; \theta)$.
- Suppose we we want to know whether $\theta = \theta_0$ or not, where θ_0 is a specific value of θ. For example, If we are testing the effect of two drugs, whose mean effects are θ_1 and θ_2, we may be interested to know if there is no difference, which corresponds to $\theta_1 - \theta_2 = 0$.
- We formalize this by stating a null hypothesis H_0 and an alternative hypothesis H_1. For example:

$$H_0 : \theta = \theta_0 \text{ versus } H_1 : \theta \neq \theta_0.$$

- More generally, consider a parameter space Θ. We consider

$$H_0 : \theta \in \Theta_0 \text{ versus } H_1 : \theta \in \Theta_1$$

where $\Theta_0 \cap \Theta_1 = \varnothing$.

Hypothesis Testing and Confidence Interval

- If Θ_0 consists of a single point, we call this a simple null hypothesis. If Θ_1 consists of more than one point, we call this a composite null hypothesis.
- Warning: Hypothesis testing should only be used when it is appropriate. Often times, people use hypothesis testing when it would be much more appropriate to use confidence intervals (which is another topic).
- Hypothesis and estimation are part of the statistical decision making, involving a loss function.

Decision Making with Minimax

- Suppose we like to estimate a parameter θ using data $X_1, \ldots, X_n$. Decision making procedure attempts to answer the following question.
- What is the best possible estimator $\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n)$ of θ.
- Minimax theory can provide a framework for answering this question.
- Loss function $L(\widehat{\theta}, \theta)$: measuring how good the estimator is
- Risk of an estimator $\widehat{\theta}$: $R(\theta, \widehat{\theta}) = E(L(\widehat{\theta}, \theta))$.
- Minimax risk:

$$R_n = inf_{\widehat{\theta}} \sup_{\theta} R(\theta, \widehat{\theta})$$

- An estimator $\widehat{\theta}$ is a minimax estimator if

$$sup_{\theta} R(\theta, \widehat{\theta}) = inf_{\widehat{\theta}} \sup_{\theta} R(\theta, \widehat{\theta})$$

Importance of Causal Inference

- The ultimate purpose of the science is to furnish causal explanation and causal relationship of observed events.
- I would rather discover one causal law than be King of Persia, —Democritus.
- We have knowledge of a thing only when we have grasped its cause; we do not have knowledge of a thing until we have grasped its why, that is to say, its cause - Aristotle